

ASSIGNMENT 1

Q1. What is Computational Design Thinking?

Q2. Differentiate between well-structured and ill-structured problems.

Q3. Explain any four characteristics of an effective problem solver.

Q4. “A college canteen has received complaints that students have to wait for a long time during lunch hours. The college wants to improve the situation.”

Answer the following:

a) Identify whether this is a well-structured or ill-structured problem. Give a reason.

b) How can Decomposition be used to analyze this problem? Give at least three sub-problems.

c) Identify one possible pattern that could be observed in this problem.

d) Mention two characteristics of an effective problem solver that would be useful in solving this problem. Explain briefly.

Q5. Explain Pattern Recognition and Algorithmic Thinking in Computational Thinking.

Q6. What is empathy mapping?

Q7. Name and explain the four major components of Computational Thinking.

Q8. Why are user interviews important?

Q9. What is a functional and technical requirement?

Q10. What are the stages of the IDEAL model? Explain with a suitable example.